

# Update Internship – Week 7

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## What we did

Removed 8 fine labels whose names match metacategories:

*floral, fruity, sweet, woody, green, spicy, citrus, earthy*

Then ran HMCN-F varying  $\beta \in [0.1, 1.0]$  and  $\lambda_{\text{viol}} \in \{0.0, 0.1, 1.0\}$

## What we found

Neither  $\beta$  nor  $\lambda_{\text{viol}}$  produce meaningful variation in performance.

Metrics remain flat across all tested values.

## Root cause

Metacategories are defined as OR over children:

$$y_{\text{meta}} = \max_{c \in \text{children}} y_c$$

Even after removing name overlaps, meta labels carry zero independent supervisory signal beyond their children.

**Learning rate** : controls step size during gradient descent:

$$\theta \leftarrow \theta - \eta \cdot \nabla_{\theta} \mathcal{L}$$

Too large: training diverges. Too small: slow convergence.

**Weight decay** : L2 regularization to prevent overfitting:

$$\mathcal{L}_{\text{reg}} = \mathcal{L} + \lambda_{\text{wd}} \|\theta\|^2$$

Penalizes large weights; encourages generalization.

$\lambda_{\text{viol}}$  : penalizes hierarchical violations in the loss:

$$\mathcal{L} = \mathcal{L}_L + \mathcal{L}_G + \lambda_{\text{viol}} \cdot \mathcal{L}_{\text{viol}}$$

where  $\mathcal{L}_{\text{viol}}$  fires when a child score exceeds its parent. Intended to enforce  $\hat{y}_{\text{child}} \leq \hat{y}_{\text{parent}}$ .

### Final prediction in HMCN-F

$$P_F = \beta \cdot \underbrace{\left[ P_L^1 \parallel \dots \parallel P_L^{|H|} \right]}_{\text{local}} + (1 - \beta) \cdot \underbrace{P_G}_{\text{global}}$$

### All metrics across $\beta$ values (12 metacategories)

| $\beta$ | ROC AUC      | PR AUC        | F1 <sub>mac</sub> | Jaccard      | Hamming      |
|---------|--------------|---------------|-------------------|--------------|--------------|
| 0.1     | 0.7924       | 0.6181        | 0.6036            | 0.4382       | 0.2380       |
| 0.3     | 0.7946       | 0.6172        | 0.6026            | 0.4361       | 0.2435       |
| 0.5     | 0.7949       | 0.6193        | 0.6030            | 0.4369       | 0.2538       |
| 0.7     | 0.7922       | 0.6204        | 0.6066            | 0.4411       | 0.2448       |
| 0.9     | 0.7916       | <b>0.6209</b> | 0.6051            | 0.4398       | 0.2418       |
| 1.0     | 0.7891       | 0.6179        | 0.5966            | 0.4305       | 0.2463       |
| Range   | <b>0.006</b> | <b>0.004</b>  | <b>0.010</b>      | <b>0.011</b> | <b>0.016</b> |

## Effect per parameter (macro avg. over 12 metacategories)

| Param                   | Value              | ROC          | PR           | F1 <sub>mac</sub> | Prec <sub>mac</sub> | Hamming      |
|-------------------------|--------------------|--------------|--------------|-------------------|---------------------|--------------|
| lr                      | $10^{-3}$          | <b>0.801</b> | <b>0.628</b> | <b>0.602</b>      | <b>0.549</b>        | <b>0.245</b> |
|                         | $5 \times 10^{-4}$ | 0.799        | 0.626        | 0.601             | 0.550               | 0.244        |
|                         | $10^{-4}$          | 0.796        | 0.625        | 0.596             | 0.538               | 0.250        |
| $\lambda_{\text{viol}}$ | 0.00               | 0.799        | 0.627        | 0.600             | 0.546               | 0.247        |
|                         | 0.01               | 0.798        | 0.625        | 0.599             | 0.546               | 0.248        |
|                         | 0.05               | 0.799        | 0.626        | 0.600             | 0.545               | 0.246        |
|                         | 0.10               | 0.798        | 0.627        | 0.599             | 0.548               | 0.246        |
| dropout                 | 0.30               | <b>0.799</b> | <b>0.628</b> | <b>0.601</b>      | <b>0.549</b>        | <b>0.245</b> |
|                         | 0.47               | 0.798        | 0.625        | 0.599             | 0.543               | 0.248        |
| wd                      | $10^{-4}$          | <b>0.799</b> | <b>0.628</b> | 0.600             | <b>0.549</b>        | <b>0.245</b> |
|                         | $10^{-3}$          | 0.798        | 0.625        | 0.600             | 0.543               | 0.248        |

## Effect per parameter (macro avg. over 138 fine labels)

| Param                   | Value              | ROC          | PR           | F1 <sub>mac</sub> | F1 <sub>mic</sub> | Prec <sub>mac</sub> | Hamming      |
|-------------------------|--------------------|--------------|--------------|-------------------|-------------------|---------------------|--------------|
| lr                      | $10^{-3}$          | <b>0.770</b> | <b>0.150</b> | <b>0.145</b>      | <b>0.246</b>      | 0.133               | 0.064        |
|                         | $5 \times 10^{-4}$ | 0.764        | 0.148        | 0.144             | 0.246             | <b>0.138</b>        | <b>0.060</b> |
|                         | $10^{-4}$          | 0.692        | 0.113        | 0.107             | 0.219             | 0.096               | 0.063        |
| $\lambda_{\text{viol}}$ | 0.00               | 0.743        | 0.135        | 0.128             | 0.236             | 0.118               | 0.062        |
|                         | 0.01               | 0.735        | 0.134        | 0.130             | 0.233             | 0.119               | 0.063        |
|                         | 0.05               | 0.742        | 0.136        | 0.133             | 0.238             | 0.122               | 0.062        |
|                         | 0.10               | 0.753        | 0.146        | 0.141             | 0.242             | 0.133               | 0.062        |
| dropout                 | 0.30               | 0.738        | 0.136        | 0.132             | 0.236             | 0.124               | 0.062        |
|                         | 0.47               | 0.748        | 0.139        | 0.133             | 0.239             | 0.121               | 0.062        |
| wd                      | $10^{-4}$          | <b>0.788</b> | <b>0.175</b> | <b>0.169</b>      | <b>0.270</b>      | <b>0.168</b>        | <b>0.051</b> |
|                         | $10^{-3}$          | 0.697        | 0.099        | 0.095             | 0.204             | 0.076               | 0.074        |

## Recommended configuration

| Parameter               | Value     |
|-------------------------|-----------|
| lr                      | $10^{-3}$ |
| weight decay            | $10^{-4}$ |
| dropout                 | 0.30      |
| $\lambda_{\text{viol}}$ | 0.0       |
| $\beta$                 | 0.5       |

## Performance at best configuration

| Metric              | Value        |
|---------------------|--------------|
| ROC AUC             | 0.809        |
| PR AUC              | <b>0.639</b> |
| F1 <sub>mac</sub>   | <b>0.609</b> |
| Prec <sub>mac</sub> | 0.571        |
| Hamming             | <b>0.233</b> |

## PR AUC (macro avg. over 12 metacategories)

| Model | Radius 1 | Radius 2     | Radius 3 |
|-------|----------|--------------|----------|
| LR    | 0.496    | 0.569        | 0.512    |
| RF    | 0.560    | <b>0.623</b> | 0.570    |
| SVM   | 0.530    | 0.593        | 0.545    |
| XGB   | 0.539    | 0.598        | 0.554    |
| KNN   | 0.452    | 0.509        | 0.450    |

## Macro-F1

| Model | Radius 1 | Radius 2     | Radius 3 |
|-------|----------|--------------|----------|
| LR    | 0.525    | 0.559        | 0.524    |
| RF    | 0.516    | 0.550        | 0.517    |
| SVM   | 0.505    | <b>0.577</b> | 0.502    |
| XGB   | 0.527    | 0.549        | 0.504    |
| KNN   | 0.491    | 0.547        | 0.500    |

## ROC AUC

| Model | Radius 1     | Radius 2 | Radius 3     |
|-------|--------------|----------|--------------|
| LR    | 0.812        | 0.810    | <b>0.816</b> |
| RF    | <b>0.827</b> | 0.821    | <b>0.828</b> |
| SVM   | 0.822        | 0.818    | 0.824        |
| XGB   | 0.807        | 0.804    | 0.813        |
| KNN   | 0.771        | 0.772    | 0.770        |

## Architecture

A RoBERTa-based transformer pretrained on 77M SMILES strings from PubChem and ZINC.

SMILES are treated as plain text, each character or subword token is fed directly into the transformer.

Produces a 768-dimensional embedding from the [CLS] token representing the full molecule.

## Pretraining objective

**Masked Language Modeling (MLM):** randomly mask tokens in the SMILES string and train the model to predict them:

$$\mathcal{L}_{\text{MLM}} = - \sum_{i \in \mathcal{M}} \log P(x_i | x_{\setminus \mathcal{M}})$$

## Limitation for our task

ChemBERTa learns to reconstruct SMILES tokens, it has no direct exposure to molecular properties during pretraining.

The link between SMILES syntax and odor perception is indirect.

## Advantage

Very large pretraining corpus (77M molecules) gives broad chemical coverage.

Easy to fine-tune: discard MLM head, attach a classification head on top of [CLS].

# What is MolBERT?

## Architecture

A BERT-based transformer pretrained on 4M molecules from ChEMBL.

Uses **Morgan atom identifiers** as tokens , each token represents a chemically meaningful substructure.

Produces a 768-dimensional embedding from the [CLS] token.

## Pretraining objectives

1. Masked Language Modeling , same as ChemBERTa.
2. Physicochemical property prediction simultaneously predicts 200 molecular properties (logP, MW, TPSA, ...)

## Advantages

Chemically meaningful tokenization via Morgan atom identifiers

Property-supervised pretraining , learns structure→property mapping

Smaller corpus but higher chemical information density per token

## Limitations

smaller pretraining corpus (4M vs 77M)

Pretrained on ChEMBL (drug-like molecules) , distribution shift from odorant molecules

More complex tokenization pipeline

## ChemBERTa vs MolBERT : Comparison

|                       | ChemBERTa                       | MolBERT                                          |
|-----------------------|---------------------------------|--------------------------------------------------|
| Base architecture     | RoBERTa                         | BERT                                             |
| Tokenization          | SMILES as plain text            | Morgan atom identifiers                          |
| Pretraining data      | 77M molecules<br>PubChem + ZINC | 4M molecules (ChEMBL)                            |
| Pretraining objective | MLM only                        | MLM + 200 physicochemical<br>property prediction |
| Chemical awareness    | Syntax only                     | Structure + properties                           |
| Embedding size        | 768-dim                         | 768-dim                                          |